

Charge Conjugation as Orientation Reversal in Hyperbolic Flavor Geometry

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Abstract

We identify the charge conjugation operation $\mathcal{C}$ with orientation reversal of the compact hyperbolic 3-manifolds underlying the hyperbolic flavor geometry framework. The lepton and quark sector manifolds $m003$ and $m006$ are both confirmed chiral—neither is isometric to its mirror image—via two independent methods: the Chern–Simons invariant, which changes sign under orientation reversal, and direct verification that `is_amphicheiral() = False` in SnapPy, with symmetry groups $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ containing no orientation-reversing elements. Under the identification $\mathcal{C} : M \mapsto \bar{M}$, the volume and first homology are preserved (same mass spectrum and generation structure), while the Chern–Simons invariant changes sign, distinguishing particle from antiparticle. For the lepton sector: the electron corresponds to $m003$ with $\text{CS} = \frac{1}{4}$ and the positron to $\bar{m}003$ with $\text{CS} = \frac{3}{4}$. The $\mathbb{Z}/5$ torsion structure admits a CP-odd phase $e^{2\pi i/5}$ that breaks the $M \leftrightarrow \bar{M}$ symmetry, providing a geometric origin for matter–antimatter asymmetry. All results are computed from census data using SnapPy [9] and are reproducible.

1 Introduction

The Standard Model of particle physics treats particles and antiparticles as distinct objects related by charge conjugation $\mathcal{C}$, but provides no geometric origin for this operation. In the hyperbolic flavor geometry framework [2, 3, 4], the flavor parameters of the Standard Model arise from the arithmetic and geometry of compact hyperbolic 3-manifolds. The CKM quark mixing matrix emerges from the holonomy of the manifold $m006 = \text{OrientableClosedCensus}[43]$ ($H_1 = \mathbb{Z}/5$, $\text{vol} = 2.029$), the PMNS lepton mixing matrix from the Meyerhoff manifold $m003 = \text{OrientableClosedCensus}[1]$ ($H_1 = \mathbb{Z}/5$, $\text{vol} = 0.981$), and CP phases from flat $U(1)$ connections classified by $H_1(M) = \mathbb{Z}/5$ [4].

A natural question arises: where do antiparticles sit in this geometric framework? In quantum field theory, charge conjugation relates a particle to its antiparticle, reversing all additive quantum numbers (charge, lepton number, baryon number) while preserving mass

and spin. The mass spectrum is preserved because particles and antiparticles have identical masses. In a geometric framework where mass is encoded in the volume and geodesic spectrum of a manifold, the natural geometric operation preserving these invariants while distinguishing particle from antiparticle is *orientation reversal*: $M \mapsto \bar{M}$, where $\bar{M}$ denotes M with its orientation reversed.

This identification is non-trivial only if the manifolds are chiral — that is, if M and $\bar{M}$ are geometrically distinct. If M were amphicheiral (isometric to its mirror image), the operation $M \mapsto \bar{M}$ would be trivial and could not distinguish particle from antiparticle. We show that both $m003$ and $m006$ are chiral, confirmed by two independent methods, making the identification $\mathcal{C} : M \mapsto \bar{M}$ geometrically meaningful.

The paper is organized as follows. Section 2 establishes the chirality of the flavor manifolds. Section 3 defines geometric charge conjugation and identifies its transformation properties. Section 4 discusses the connection to matter–antimatter asymmetry. Section 5 states falsifiable predictions. Section 6 concludes.

2 Chirality of the flavor manifolds

A compact hyperbolic 3-manifold M is *chiral* if it is not isometric to its orientation-reversed mirror image $\bar{M}$; it is *amphicheiral* if it is isometric to $\bar{M}$.

2.1 The Chern–Simons invariant as a chirality detector

The Chern–Simons invariant $\text{CS}(M) \in \mathbb{R}/\mathbb{Z}$ is a topological invariant of oriented hyperbolic 3-manifolds that transforms as

$$\text{CS}(\bar{M}) = -\text{CS}(M) \pmod{1} \quad (1)$$

under orientation reversal [5]. If $\text{CS}(M) \not\equiv 0$ and $\text{CS}(M) \not\equiv \frac{1}{2} \pmod{1}$, then $\text{CS}(M) \neq \text{CS}(\bar{M})$ and M cannot be isometric to $\bar{M}$: the manifold is chiral.

For the lepton sector manifold $m003$:

$$\text{CS}(m003) = \frac{1}{4}, \quad \text{CS}(\overline{m003}) = -\frac{1}{4} \equiv \frac{3}{4} \pmod{1}. \quad (2)$$

Since $\frac{1}{4} \neq \frac{3}{4}$, $m003$ and $\overline{m003}$ are geometrically distinct: $m003$ is chiral.

For the quark sector manifold $m006$:

$$\text{CS}(m006) = -0.114137, \quad \text{CS}(\overline{m006}) = +0.114137. \quad (3)$$

Since these values differ, $m006$ is chiral. (The value for $m006$ is given to six decimal places; it is conjectured to be irrational, consistent with the non-arithmetic status of $m006$.)

2.2 Direct verification via SnapPy

Independent confirmation is obtained from the symmetry group of each manifold. A compact hyperbolic 3-manifold is amphicheiral if and only if its isometry group contains orientation-reversing elements. Using SnapPy [9]:

Manifold	is_amphicheiral()	Symmetry group	Order
$m003$ (OCC[1])	False	$\mathbb{Z}/2 \oplus \mathbb{Z}/2$	4
$m006$ (OCC[43])	False	$\mathbb{Z}/2 \oplus \mathbb{Z}/2$	4

Both symmetry groups have order 4 and consist entirely of orientation-preserving isometries. No orientation-reversing element exists, confirming chirality independently of the Chern–Simons computation. The two methods agree: both $m003$ and $m006$ are chiral.

3 Geometric charge conjugation

Definition 1 (Geometric charge conjugation). In the hyperbolic flavor geometry framework, charge conjugation $\mathcal{C}$ acts as orientation reversal on the underlying manifold:

$$\mathcal{C} : M \mapsto \bar{M}.$$

The naturality of this definition rests on the transformation properties of the geometric invariants that encode physical observables:

Proposition 2 (Transformation properties of $\mathcal{C}$). *Under $\mathcal{C} : M \mapsto \bar{M}$:*

$$\text{vol}(\bar{M}) = \text{vol}(M), \quad (4)$$

$$H_1(\bar{M}) \cong H_1(M), \quad (5)$$

$$\text{CS}(\bar{M}) = -\text{CS}(M) \pmod{1}. \quad (6)$$

Proof. Equation (4): hyperbolic volume is an orientation-independent invariant of the Riemannian metric. Equation (5): first homology depends only on the topological type, not the orientation. Equation (6): see [5]. $\square$

Corollary 3 (Physical interpretation). *Under $\mathcal{C}$: the mass spectrum is preserved (vol encodes mass scale), the generation structure is preserved ($H_1 = \mathbb{Z}/5$ encodes three generations), and the Chern–Simons invariant changes sign, distinguishing particle from antiparticle.*

3.1 Particle–antiparticle assignments

The identification yields concrete geometric assignments for the elementary fermions and their antiparticles:

Sector	Particle / Antiparticle	Manifold	CS
Lepton	Electron e^-	$m003$	$+\frac{1}{4}$
	Positron e^+	$\overline{m003}$	$+\frac{3}{4}$
Quark	Quark q	$m006$	-0.1141
	Antiquark $\bar{q}$	$\overline{m006}$	$+0.1141$

The rational value $\text{CS}(e^-) = \frac{1}{4}$ for the electron reflects the arithmetic structure of $m003$: arithmetic hyperbolic 3-manifolds have rational Chern–Simons invariants (up to integer shifts), while non-arithmetic ones generally have irrational CS [6]. The irrational value $\text{CS}(q) = -0.1141$ for the quark is consistent with the conjectured non-arithmetic status of $m006$.

3.2 Consistency with CPT

In quantum field theory, CPT is an exact symmetry. In the geometric framework, the three discrete operations act as:

- $\mathcal{C}$: orientation reversal $M \rightarrow \bar{M}$,
- $\mathcal{P}$: spatial reflection (acts on the embedding space, not the internal manifold),
- $\mathcal{T}$: time reversal, equivalent to $\mathcal{CP}$ by the CPT theorem, i.e., orientation reversal combined with spatial reflection.

The combination $\mathcal{CPT}$ acts as the identity on the manifold (two orientation reversals), consistent with CPT invariance.

4 Matter–antimatter asymmetry

If $\mathcal{C}$ is orientation reversal, then a universe containing matter but not antimatter corresponds geometrically to a preferred orientation of the vacuum manifold. The $\mathbb{Z}/5$ torsion structure of $m003$ and $m006$ provides the mechanism for this preference.

The flat $U(1)$ connections on M are classified by $H^1(M, U(1)) \cong \text{Hom}(\mathbb{Z}/5, U(1))$, giving holonomies $e^{2\pi i k/5}$ for $k = 0, 1, 2, 3, 4$. The non-trivial character $k = 1$ gives a CP-odd phase $e^{2\pi i/5}$. Under orientation reversal, this phase transforms as

$$e^{2\pi i/5} \mapsto e^{-2\pi i/5},$$

breaking the $M \leftrightarrow \bar{M}$ symmetry. This is the geometric origin of CP violation in the lepton sector, and by extension, a candidate mechanism for leptogenesis [7]: the CP-odd phase $e^{2\pi i/5}$ generates a lepton asymmetry that is converted to a baryon asymmetry via electroweak sphalerons.

The $\mathbb{Z}/5$ torsion yields a Jarlskog invariant $J = \frac{1}{4} \sin(2\pi/5) \approx 0.0309$, which matches the PDG value $J = 0.033 \pm 0.003$ to within 6% (see companion paper [4]). The actual PMNS CP phase δ depends on the parameterization; the geometric origin implies $|\sin \delta| \approx 1$, i.e., $\delta \approx 90^\circ$ or 270° . The PDG best fit for normal ordering is $\delta \approx 197^\circ$, which is within 27° of $180^\circ + 72^\circ = 252^\circ$; the precise mapping requires a full reparameterisation of the geometric matrix.

It is important to note that the present framework provides a geometric origin for the *existence* of a CP-odd phase but does not dynamically explain why the universe selected the orientation M rather than $\bar{M}$. A dynamical mechanism for orientation selection remains an open problem.

¹Chern–Simons invariants are taken modulo 1 in the range $[0, 1)$. Hence $-\frac{1}{4} \equiv \frac{3}{4}$.

5 Falsifiable predictions

The geometric identification $\mathcal{C} : M \mapsto \bar{M}$ yields the following predictions:

Chern–Simons invariants. If the Chern–Simons invariant has physical significance (as in Chern–Simons TQFT approaches to quantum gravity [8]), the electron sector is characterized by $\text{CS} = \frac{1}{4}$ and the positron sector by $\text{CS} = \frac{3}{4}$. Any future geometric derivation of electric charge from 3D topology must reproduce this assignment.

Arithmetic vs non-arithmetic sectors. The rational $\text{CS}(m003) = \frac{1}{4}$ implies $m003$ is arithmetic [6], while the irrational $\text{CS}(m006) = -0.1141$ implies $m006$ may be non-arithmetic. If $m006$ is proven arithmetic, its CS invariant would be rational and the current value would require reinterpretation.

No new species prediction. The framework assigns a unique manifold to each fermion generation via the $\mathbb{Z}/5$ homology structure. Additional fermion generations would require manifolds with $H_1 \cong \mathbb{Z}/p$ for $p > 5$ or more complex torsion structures. The absence of a fourth generation in the closed census with appropriate properties is consistent with three generations being the complete set.

CP violation magnitude. The Jarlskog invariant $J = \frac{1}{4} \sin \frac{2\pi}{5} \approx 0.031$ is fixed by the $\mathbb{Z}/5$ torsion structure. If future measurements shift the PDG value $J = 0.033 \pm 0.003$ by more than 3σ away from 0.031, the geometric CP mechanism would be disfavored.

6 Conclusion

We have established that the two compact hyperbolic 3-manifolds underlying the hyperbolic flavor geometry framework — $m003$ (lepton sector) and $m006$ (quark sector) — are both chiral, confirmed by the Chern–Simons invariant and by direct SnapPy computation. This chirality makes the identification of charge conjugation $\mathcal{C}$ with orientation reversal $M \mapsto \bar{M}$ geometrically non-trivial and meaningful. Under this identification, the volume and first homology (mass spectrum and generation structure) are preserved, while the Chern–Simons invariant changes sign, distinguishing particle from antiparticle. The electron corresponds to $m003$ with $\text{CS} = \frac{1}{4}$; the positron corresponds to $\overline{m003}$ with $\text{CS} = \frac{3}{4}$.

The $\mathbb{Z}/5$ torsion structure provides a CP-odd phase $e^{2\pi i/5}$ that breaks the $M \leftrightarrow \bar{M}$ symmetry, offering a geometric origin for the matter–antimatter asymmetry of the universe. The Jarlskog invariant $J \approx 0.031$ derived from this phase matches the PDG value to within 6%.

The identification presented here is a natural consequence of the chirality of the flavor manifolds and the general principle that orientation reversal is the discrete geometric operation that preserves metric invariants while reversing orientation-dependent quantities. Whether this geometric structure reflects a deeper physical principle or an emergent pattern remains an open question for future work.

Data Availability

All computations use SnapPy [9] applied to the `OrientableClosedCensus`. Scripts are available at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

Conflict of Interest

The author declares no conflicts of interest.

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